

Problem Statement

Hospital Patient Management System using Linked List

Develop a simple C++ Hospital Patient Management System using a singly linked list. The system should allow the user to:

Add a new patient with Patient ID and Name.

Display all patient records.

Search for a patient using Patient ID.

Exit the system.

The patient records are stored using nodes of a linked list, where each node contains the patient ID, patient name, and a pointer to the next patient.

Objective

- 1.To develop a simple Hospital Patient Management System using C++.
- 2.To implement a singly linked list for storing patient records.
- 3.To add and store patient details such as ID and name.
- 4.To display all patient records efficiently.
- 5.To search for a patient using their Patient ID.

Algorithm

Step 1: Start the program.

Step 2: Set head = NULL.

Step 3: Display the menu:

- Add Patient
- Display Patients
- Search Patient
- Exit

Step 4: Read the user's choice.

Step 5: If choice is 1, create a new patient node, enter ID and name, and add it to the linked list.

Step 6: If choice is 2, traverse the linked list and display all patient records.

Step 7: If choice is 3, enter the patient ID and search the linked list.

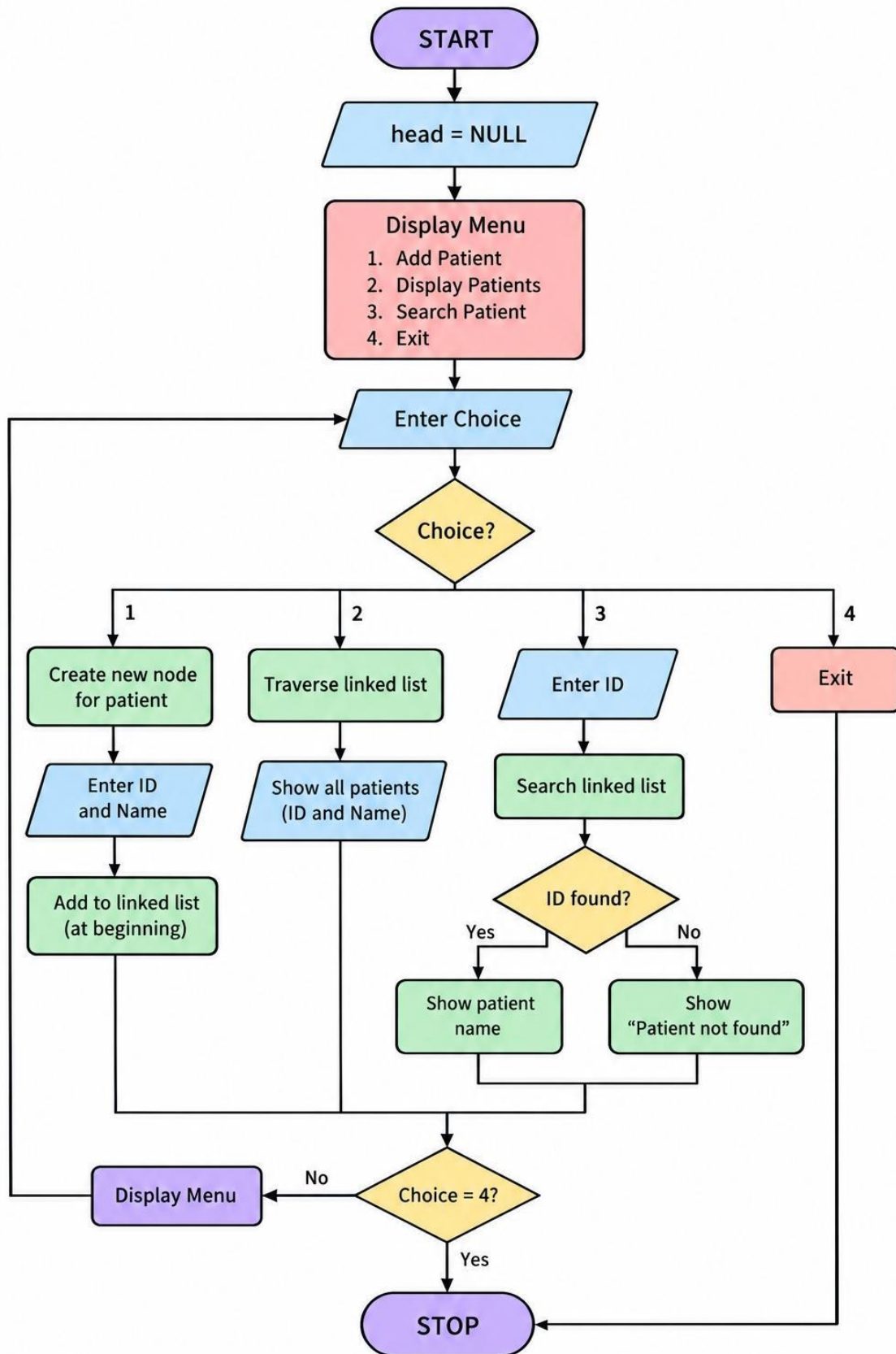
If ID is found → display patient name.

Otherwise → display “Patient not found.”

Step 8: If choice is 4, exit the program.

Step 9: Repeat the menu until the user selects Exit.

Step 10: Stop.



Flowchart

```
#include <iostream>

using namespace std;

struct Patient {
    int id;
    string name;
    Patient *next;
};

Patient *head = NULL;

// Add Patient
void addPatient() {
    Patient *p = new Patient;

    cout << "Enter ID: ";
    cin >> p->id;
    cout << "Enter Name: ";
    cin >> p->name;

    p->next = head;
    head = p;

    cout << "Patient added.\n";
}
```

// Display Patients

void display() {

Patient *temp = head;

while (temp != NULL) {

cout << "\nID: " << temp->id;

cout << "\nName: " << temp->name;

temp = temp->next;

}

}

// Search Patient

void search() {

int id;

cout << "Enter ID to search: ";

cin >> id;

Patient *temp = head;

while (temp != NULL) {

if (temp->id == id) {

cout << "Patient Found: " << temp->name;

return;

}

temp = temp->next;

}

```
        cout << "Patient not found.";
    }

int main() {
    int choice;

    do {
        cout << "\n\n1. Add Patient";
        cout << "\n2. Display Patients";
        cout << "\n3. Search Patient";
        cout << "\n4. Exit";
        cout << "\nEnter choice: ";
        cin >> choice;

        switch(choice) {
            case 1: addPatient(); break;
            case 2: display(); break;
            case 3: search(); break;
            case 4: cout << "Exit"; break;
            default: cout << "Invalid choice";
        }

    } while(choice != 4);

    return 0;
```

}

Out put:

1. Add Patient

2. Display Patients

3. Search Patient

4. Exit

Enter choice: 1

Enter ID: 123

Enter Name: shubham

Patient added.

1. Add Patient

2. Display Patients

3. Search Patient

4. Exit

Enter choice: 1

Enter ID: 187

Enter Name: samiksha

Patient added.

1. Add Patient

2. Display Patients

3. Search Patient

4. Exit

Enter choice: 1

Enter ID: 764

Enter Name: sneha

Patient added.

- 1. Add Patient**
- 2. 2.Display Patients**
- 3. 3.Search Patient**
- 4. 4.Exit**

Enter choice: 2

ID: 764

Name: sneha

ID: 187

Name: samiksha

ID: 123

Name: shubham

- 1. Add Patient**
- 2. 2.Display Patient**
- 3. 3.Search Patient**
- 4. 4.Exit**

Enter choice: 3

Enter ID to search: 187

Patient Found: samiksha

1. Add Patient

2. Display Patients

3. Search Patient

4. Exit

Enter choice: